

The edges, faces, and facets of the polytope

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The purpose of this file is to study the border of the polytope in four-dimensional (x, y, q, t) , space defined by the inequalities

$$x \leq 0, \tag{1a}$$

$$x \geq -\frac{1}{4}, \tag{1b}$$

$$3x + y + 3q \geq 0, \tag{1c}$$

$$q \leq \frac{1}{3}, \tag{1d}$$

$$t \leq -y, \tag{1e}$$

$$t \geq y. \tag{1f}$$

Since these inequalities are linear in x, y, q , and t they define a polytope.

A four-dimensional polytope is bounded by

- n_V vertices, each of them given by four equations;
- n_E (one-dimensional) edges, each of them given by three equations and uniting two, and only two, vertices;
- n_P (two-dimensional) faces, each of them given by two equations and containing three or more vertices;
- n_F three-dimensional ‘facets’, each of them given by one equation and containing four or more vertices.

Euler’s relation for the boundary of a four-dimensional polytope is

$$n_V - n_E + n_P - n_F = 0. \tag{2}$$

In order to define the boundary of the polytope in (x, y, q, t) orbit space, we transform inequalities (1) into equations:

$$x = 0, \tag{3a}$$

$$x = -\frac{1}{4}, \tag{3b}$$

$$3x + y + 3q = 0, \tag{3c}$$

$$q = \frac{1}{3}, \tag{3d}$$

$$t = -y, \tag{3e}$$

$$t = y. \tag{3f}$$

Notice that Eqs. (3a) and (3b) are contradictory, so we cannot assume them to hold simultaneously. Moreover, Eqs. (3c), (3d), (3e), and (3f) together imply $x = -1/3$, which is outside the range $-1/4 \leq x \leq 0$ defined by Eqs. (3a) and (3b); therefore, a vertex defined by Eqs. (3c), (3d), (3e), and (3f) is not possible.

Given these two constraints, it is easy to see that the polytope has $n_V = 8$, $n_E = 16$, $n_P = 14$, and $n_F = 6$, which satisfy Eq. (2).

The eight vertices are:

$$V_6 = \left(0, -1, \frac{1}{3}, 1\right) \text{ satisfies Eqs. (3a), (3c), (3d), (3e);} \quad (4a)$$

$$V_8 = \left(0, -1, \frac{1}{3}, -1\right) \text{ satisfies Eqs. (3a), (3c), (3d), (3f);} \quad (4b)$$

$$V_1 = (0, 0, 0, 0) \text{ satisfies Eqs. (3a), (3c), (3e), (3f);} \quad (4c)$$

$$V_2 = \left(0, 0, \frac{1}{3}, 0\right) \text{ satisfies Eqs. (3a), (3d), (3e), (3f);} \quad (4d)$$

$$V_5 = \left(-\frac{1}{4}, -\frac{1}{4}, \frac{1}{3}, \frac{1}{4}\right) \text{ satisfies Eqs. (3b), (3c), (3d), (3e);} \quad (4e)$$

$$V_7 = \left(-\frac{1}{4}, -\frac{1}{4}, \frac{1}{3}, -\frac{1}{4}\right) \text{ satisfies Eqs. (3b), (3c), (3d), (3f);} \quad (4f)$$

$$V_4 = \left(-\frac{1}{4}, 0, \frac{1}{4}, 0\right) \text{ satisfies Eqs. (3b), (3c), (3e), (3f);} \quad (4g)$$

$$V_3 = \left(-\frac{1}{4}, 0, \frac{1}{3}, 0\right) \text{ satisfies Eqs. (3b), (3d), (3e), (3f).} \quad (4h)$$

The 16 edges are given, in terms of the vertices that are their extreme points, by:

$$E_1 = [V_6, V_8] \text{ satisfies Eqs. (3a), (3c), (3d);} \quad (5a)$$

$$E_2 = [V_1, V_6] \text{ satisfies Eqs. (3a), (3c), (3e);} \quad (5b)$$

$$E_3 = [V_1, V_8] \text{ satisfies Eqs. (3a), (3c), (3f);} \quad (5c)$$

$$E_4 = [V_2, V_6] \text{ satisfies Eqs. (3a), (3d), (3e);} \quad (5d)$$

$$E_5 = [V_2, V_8] \text{ satisfies Eqs. (3a), (3d), (3f);} \quad (5e)$$

$$E_6 = [V_1, V_2] \text{ satisfies Eqs. (3a), (3e), (3f);} \quad (5f)$$

$$E_7 = [V_5, V_7] \text{ satisfies Eqs. (3b), (3c), (3d);} \quad (5g)$$

$$E_8 = [V_4, V_5] \text{ satisfies Eqs. (3b), (3c), (3e);} \quad (5h)$$

$$E_9 = [V_4, V_7] \text{ satisfies Eqs. (3b), (3c), (3f);} \quad (5i)$$

$$E_{10} = [V_3, V_5] \text{ satisfies Eqs. (3b), (3d), (3e);} \quad (5j)$$

$$E_{11} = [V_3, V_7] \text{ satisfies Eqs. (3b), (3d), (3f);} \quad (5k)$$

$$E_{12} = [V_3, V_4] \text{ satisfies Eqs. (3b), (3e), (3f);} \quad (5l)$$

$$E_{13} = [V_5, V_6] \text{ satisfies Eqs. (3c), (3d), (3e);} \quad (5m)$$

$$E_{14} = [V_7, V_8] \text{ satisfies Eqs. (3c), (3d), (3f);} \quad (5n)$$

$$E_{15} = [V_1, V_4] \text{ satisfies Eqs. (3c), (3e), (3f);} \quad (5o)$$

$$E_{16} = [V_2, V_3] \text{ satisfies Eqs. (3d), (3e), (3f).} \quad (5p)$$

The 14 faces, given by the—either three or four—vertices that they encompass, are:

$$F_1 = [V_1, V_6, V_8] \text{ satisfies Eqs. (3a), (3c);} \quad (6a)$$

$$F_2 = [V_2, V_6, V_8] \text{ satisfies Eqs. (3a), (3d);} \quad (6b)$$

$$F_3 = [V_1, V_2, V_6] \text{ satisfies Eqs. (3a), (3e);} \quad (6c)$$

$$F_4 = [V_1, V_2, V_8] \text{ satisfies Eqs. (3a), (3f);} \quad (6d)$$

$$F_5 = [V_4, V_5, V_7] \text{ satisfies Eqs. (3b), (3c);} \quad (6e)$$

$$F_6 = [V_3, V_5, V_7] \text{ satisfies Eqs. (3b), (3d);} \quad (6f)$$

$$F_7 = [V_3, V_4, V_5] \text{ satisfies Eqs. (3b), (3e);} \quad (6g)$$

$$F_8 = [V_3, V_4, V_7] \text{ satisfies Eqs. (3b), (3f);} \quad (6h)$$

$$F_9 = [V_5, V_6, V_7, V_8] \text{ satisfies Eqs. (3c), (3d);} \quad (6i)$$

$$F_{10} = [V_1, V_4, V_5, V_6] \text{ satisfies Eqs. (3c), (3e);} \quad (6j)$$

$$F_{11} = [V_1, V_4, V_7, V_8] \text{ satisfies Eqs. (3c), (3f);} \quad (6k)$$

$$F_{12} = [V_2, V_3, V_5, V_6] \text{ satisfies Eqs. (3d), (3e);} \quad (6l)$$

$$F_{13} = [V_2, V_3, V_7, V_8] \text{ satisfies Eqs. (3d), (3f);} \quad (6m)$$

$$F_{14} = [V_1, V_2, V_3, V_4] \text{ satisfies Eqs. (3e), (3f).} \quad (6n)$$

The six facets $H_{1,\dots,6}$ are defined, both by the—either four or six—vertices they comprise, and by the equations that they satisfy, as:

$$H_1 = [V_1, V_2, V_6, V_8] \text{ satisfies Eq. (3a);} \quad (7a)$$

$$H_2 = [V_3, V_4, V_5, V_7] \text{ satisfies Eq. (3b);} \quad (7b)$$

$$H_3 = [V_1, V_4, V_5, V_6, V_7, V_8] \text{ satisfies Eq. (3c);} \quad (7c)$$

$$H_4 = [V_2, V_3, V_5, V_6, V_7, V_8] \text{ satisfies Eq. (3d);} \quad (7d)$$

$$H_5 = [V_1, V_2, V_3, V_4, V_5, V_6] \text{ satisfies Eq. (3e);} \quad (7e)$$

$$H_6 = [V_1, V_2, V_3, V_4, V_7, V_8] \text{ satisfies Eq. (3f).} \quad (7f)$$